

Technical Design Document (TDD)

F1tenth PoliMI

The car: stack, car interface and deployment

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Document History

Version	Date	Author	Changes
0.01	2026-10-25	F1tenth PoliMI team	Initial draft from the car-interface design proposal and the runbooks; `car.md` and the hardware studies folded in (section A)
0.02	2026-11-08	F1tenth PoliMI team	Review of v0.01: DEC-08 follows the Jazzy migration; DEC-12 allows our packages in a fork's own folder; JetPack 7.2.1 on the car (DEC-13); Jazzy migration summarised (section 7.3); stop-test chain, steering limits (T-12) and the `/joy` stand-in owner; references to working notes removed
0.03	2026-11-09	F1tenth PoliMI team	Controllers and tuning topics (PID and Pure Pursuit, FR-PTH-009); camera calibration procedure in the car bringup (FR-DRV-005, T-06); line detection on replayed camera bags; IMU model deferred until OI-08

Document Approvals

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1 Introduction

1.1 Purpose

This document defines how the car's software is built: the layers of the stack, the car interface the stack talks to, how the stack is started, where the car's measured numbers live, and how it is deployed on the car and on a developer's machine.

1.2 Scope

- **In scope:** the stack on the car, the car interface (topics and frames), the car's hardware layer, the repositories, the Docker images, the development standards.
- **Out of scope:** the simulator's internals (the gym, the bridge, `polimi_sim`). Here the simulator appears only as the second implementation of the car interface (`hw:=sim`). Its design is in `TDD_Gym`.

1.3 References

- `RASD_F1tenth` — requirements refined here.
- The README (first-time setup), `CONTRIBUTING.md` (git workflow, access, rulesets) and the runbooks `docs/software/docker.md` and `simulator.md`.
- `TDD_Gym`, the simulator's design.
- Hardware studies: LiDAR (23 Sep 2026), camera (7 Oct 2026), chassis (24 Sep 2026), folded into section A.

1.4 Status Conventions

[Decided] agreed **[Proposed]** proposed, waiting for the team's confirmation **[V2]** later phase **[TBD]** open.

> **Today versus target.** The images run ROS 2 Humble today and the drivers come from `humble-polimi`. Jazzy is decided (DEC-01) and the migration is planned (section 7.3), not done. This document describes the Jazzy target; where today differs, it says so.

2 Architecture

> **Key point.** One stack, two cars. The stack talks to the car only through a fixed set of ROS topics and TF frames, the **car interface**. On the car, `f1tenth_system`'s drivers and `orbbec_camera` serve it. In the simulator, our fork of `f1tenth_gym_ros` serves the same interface. The stack cannot tell which one it is running on, and neither fork depends on anything of ours. **[Proposed]**

2.1 Component View

Three layers. The logic and the common layer are one stack and run unchanged everywhere. The hardware layer is exactly one of two, picked by the launch argument `hw:=car` or `hw:=sim`.

The seam sits below the mux, at `/ackermann_cmd`, the topic the VESC reads. The RASD requires joystick takeover, the deadman and AEB to work on the car and in the simulator (FR-SIM-002), so the mux and the safety nodes belong to the common layer, not to the hardware.

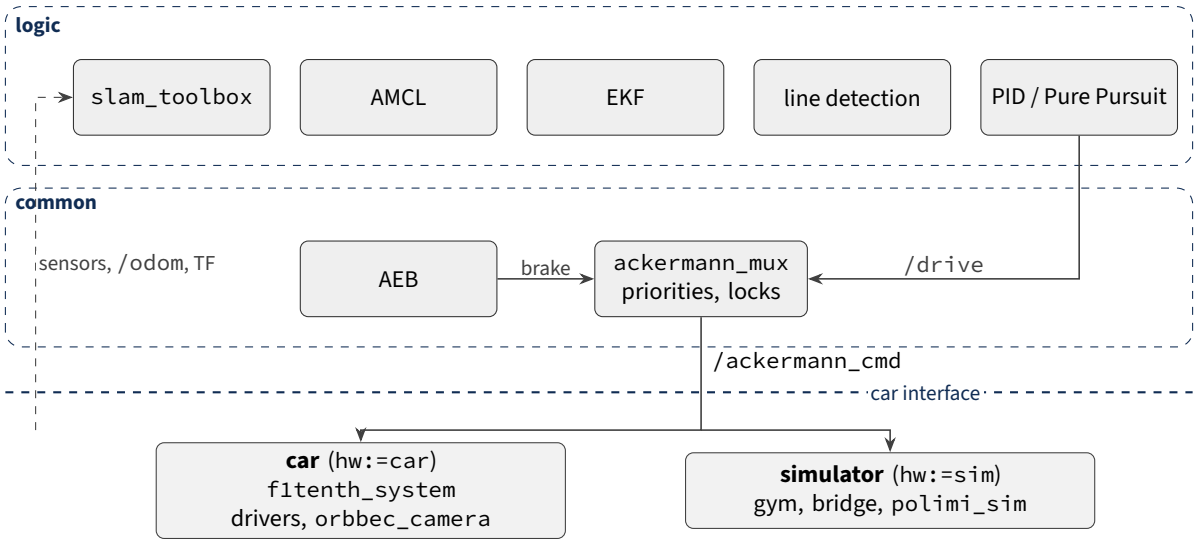


Figure 1: Layers. Only `/ackermann_cmd` goes down through the car interface; sensors, odometry and TF come up with the same names whichever hardware runs.

2.2 Components

Component	Responsibility	Interfaces
Logic		
`slam_toolbox`	Map the room (FR-MAP)	`/scan`, TF → `/map`
AMCL	Pose in the room map (FR-LOC)	`/scan`, `/odom`, `/map` → TF `map`→`odom`
EKF (`robot_localization`)	Fused `odom` **[TBD: OI-08]**	`/odom`, IMU → TF `odom`→`base_link`
`polimi_lines`	Line detection (FR-LIN)	camera image, info → `nav_msgs/Path`
`polimi_control`	PID and Pure Pursuit, same interface (FR-PTH)	path, TF → `/drive`, tracking errors
Common		
`ackermann_mux`	Pick the highest-priority input still sending; locks (FR-SAF)	`/teleop`, `/drive`, `/joy` → `/ackermann_cmd`
`polimi_safety`	AEB (FR-SAF-004)	`/scan`, `/odom` → mux input
`polimi_bringup`	The car interface (README), launches, our configs	`stack.launch.py hw:=car|sim`
Hardware, car		
`vesc_driver`, `vesc_ackermann`	Motor, servo, odometry (FR-DRV-001)	`/ackermann_cmd` → VESC; VESC → `/odom`
`joy`, `joy_teleop`	Joystick (FR-DRV-002)	`/joy`, `/teleop`
`orbbec_camera`	SL450 LiDAR, Gemini 335L camera and IMU (FR-DRV-003)	`/scan`, image, info, IMU
Hardware, simulator (see `TDD_Gym`)		

Component	Responsibility	Interfaces
fork of `f1tenth_gym_ros`	Serves the same interface	as the car

2.3 Runtime View

2.3.1 Command path

Autonomy publishes `/drive`. A person drives through `/teleop`, from the joystick on the car or the keyboard in the simulator. The mux picks by priority (joystick 100, autonomy 10, as in today's `mux.yaml`); the safety locks of OI-09 sit in the same mux. Below the seam, the car turns `/ackermann_cmd` into VESC commands; the simulator turns it into gym actions through its actuation model.

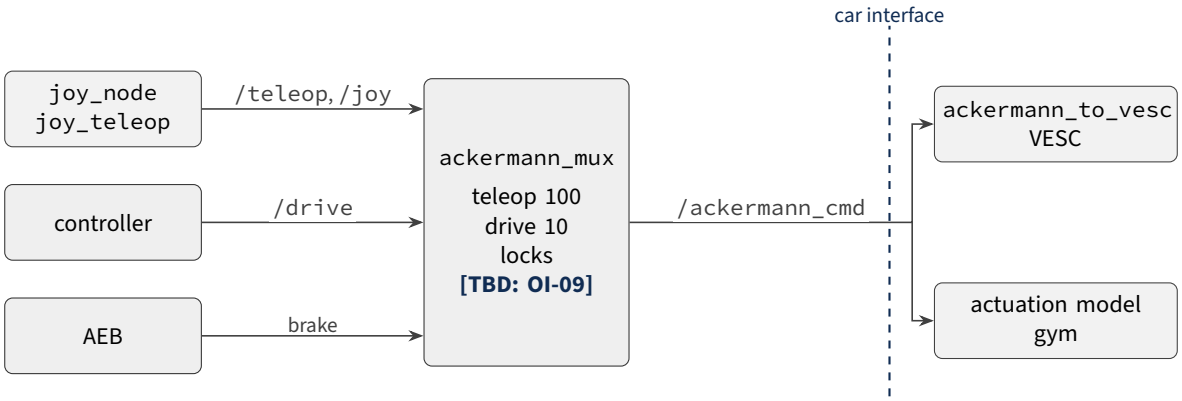


Figure 2: Command path. Grey boxes are hardware-specific: in the simulator a keyboard bridge publishes `/teleop` in place of the joystick nodes, and the car's VESC is replaced by the actuation model. Everything else is identical on car and simulator, so takeover, deadman and AEB are tested in the simulator as they run on the car.

2.3.2 Sensor path

Each topic in the middle column is part of the car interface. The car and the simulator each have one producer per topic; the stack subscribes only to the middle column. The simulator's ground truth stays out of it: it is for debugging and scoring only, never read by the stack (FR-SIM-003).

2.4 Deployment View

2.4.1 Machines and images

One Docker image per machine type, built from `docker/ros/Dockerfile`. The repo's code, configs and maps are bind-mounted, so code is edited on the host and run in the container; `build/`, `install/` and `log/` live in Docker volumes.

Service	Machine	Simulator	Networking
`arm`	Apple Silicon Mac	yes	port 8765 (Foxglove), bound to 127.0.0.1
`x86`	Linux or Windows PC	yes	port 8765 (Foxglove), bound to 127.0.0.1

`jet`	Jetson Orin Nano on the car	no	host
`novnc`	optional, RViz in the browser	–	port 8080

Base image `ros:jazzy` on Ubuntu 24.04 (DEC-01); today `ros:humble`. On the car the host is JetPack 7.2.1, also Ubuntu 24.04, so the container can use the GPU (DEC-13). Middleware `rmw_zenoh_cpp`; the Zenoh router starts with the container.

2.4.2 The car

Device	Link to the Orin Nano	Driver
Orbbec Pulsar SL450	Ethernet (the only port)	`orbbec_camera`, `lidar.launch.py`
Orbbec Gemini 335L	USB 3	`orbbec_camera`, `gemini_330_series.launch.py`
VESC 6 MkVI	USB, `/dev/sensors/vesc` (udev rule **[TBD]**)	`vesc_driver`
8BitDo Ultimate 2C	wireless dongle, USB	`joy`

2.4.3 Repositories

Repo (`tfc-org`)	Submodule path	Holds
`f1tenth_polimi`	– (this repo)	Our packages, configs, maps, Docker, docs
`f1tenth_system_polimi`	`ros2_ws/src/f1tenth_system`	Upstream drivers, branch `jazzy-polimi` (today `humble-polimi`)
`f1tenth_gym_polimi`	`ros2_ws/src/f1tenth_gym_ros` (after the migration)	Upstream bridge plus `polimi_sim`, branch `dev-jazzy-polimi`

Dependencies point one way only: from our repo outwards. Neither fork knows our stack exists (DEC-08).

2.4.4 Launch

One entry point, `stack.launch.py`, with `hw:=car` or `hw:=sim`. It always starts the common layer and the logic asked for, plus exactly one hardware launch. **[Proposed]**

- `stack.launch.py hw:=car|sim`
  - `common.launch.py`: `ackermann_mux` with our `mux.yaml`, safety.
  - `car.launch.py` (`hw:=car`): `f1tenth_system`'s driver nodes one by one, `orbbec_camera`, the mount TF from `config/car/`.
  - `sim.launch.py` (`hw:=sim`): includes the fork's `sim_car.launch.py` with `config_dir:=~/ws/config` and, from DEC-08, `teleop_topic:=teleop`; it forwards `camera_source` and `camera_uri`; nothing else.

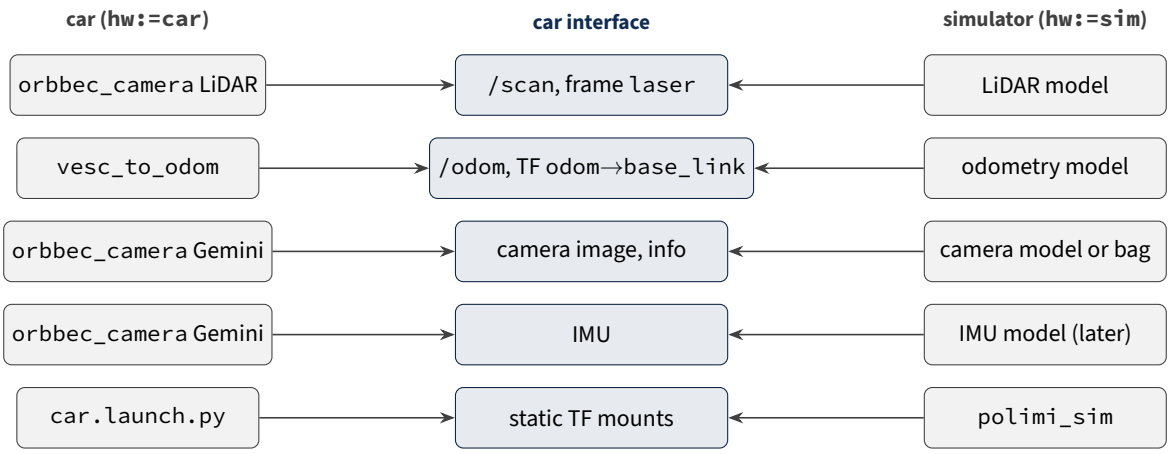


Figure 3: Sensor path. Same topic, same type, same frame, two producers. The mount TF comes from the same measured files on both sides (DEC-09), so the simulator places the LiDAR where the car's TF says it is.

- bag replay (`hw:=sim camera_source:=bag camera_uri:=<bag>`, `TDD_Gym`): every stack node gets `use_sim_time:=true`, so TF lookups run on the bag's clock, and only line detection runs.
- the logic: SLAM or localization, line detection, controller, as asked.

Today the car starts with `f1tenth_stack`'s `no_lidar_bringup_launch.py` and the simulator with `simcar`, which becomes a shortcut for `hw:=sim`.

2.5 Design Decisions

ID	Decision	Rationale / alternatives
DEC-01	ROS 2 Jazzy on every image **[Decided]**	Humble ends on 31 May 2027, before or during the summer school; Jazzy is supported to 31 May 2029. Lyrical is too new: no `f1tenth_gym_ros` branch, no JetPack for its Ubuntu. (OI-13, 21 Oct 2026.)
DEC-02	One image rebuild adds `nav2_amcl`, `robot_localization`, `image_transport_plugins`, `cv_bridge` **[Decided]**	In the Jazzy base apt layer, so AMCL and the EKF can be tried before any stack package exists. `cv_bridge` brings apt's `python3-opencv`; the apt layer goes first and pip after, so the gym's pip `cv2` and NumPy shadow apt's. Python nodes never import `cv_bridge` (its Python module breaks under the gym's NumPy 2) and convert images with NumPy; C++ nodes are unaffected. A `numpy<2` pin is no fallback, as the JAX gym needs NumPy $\geq$ 2.2: if a Python node ever needs `cv_bridge`, build it from source against NumPy 2. Checked after the rebuild (section 7.3, step 1). (OI-10, 22 Oct 2026.)
DEC-03	Seam below the mux, at `/ackermann_cmd` **[Proposed]**	The simulator replaces the VESC, the motor, the chassis and the sensors, nothing above them. With the mux above the seam, takeover, deadman and AEB are the same nodes in both places. (R1)

ID	Decision	Rationale / alternatives
DEC-04	Plain topics are the contract; no extension mechanism **[Proposed]**	Our launch includes the fork's launch with two arguments; one switch, `hw:=`. No new code in the fork, and the contract can be checked with `ros2 topic info` on both sides. (R2)
DEC-05	`polimi_bringup` holds the contract and the launches; one package per function for the logic **[Proposed]**	Functions have different dependencies (the camera stack versus none) and their own tests; one bringup package keeps how it all starts in one place. (R3)
DEC-06	Our own `car.launch.py`, not `f1tenth_stack`'s bringup **[Proposed]**	Upstream's `bringup_launch.py` starts the Hokuyo driver, its own mux and a hard-coded laser TF, none of which match our car. `f1tenth_system` stays a pure upstream fork: the scope's "our fork, extended with all our drivers" (`RASD_F1tenth` C-022) becomes "its driver packages, launched by our bringup". (R4)
DEC-07	Our own `mux.yaml`, in `polimi_bringup/config/` **[Proposed]**	Starts from today's values (joystick 100, autonomy 10). OI-09 adds the e-stop and deadman locks and the AEB input, and those are ours, not upstream's. (R5)
DEC-08	The simulator fork runs from a bare clone, after the Jazzy migration **[Proposed]**	During the migration the fork keeps `ackermann_mux` in `sim_car.launch.py` and needs `f1tenth_system` next to it, so the migration changes only the distro, the gym and the repo layout (8 Nov 2026). Afterwards, with `polimi_bringup`'s `common.launch.py`: `polimi_sim` drops the mux node and its `ackermann_mux` and `f1tenth_stack` dependencies, and gains `teleop_topic` (default `ackermann_cmd`). Earlier, the simulator would have no mux until the common layer exists. (R6)
DEC-09	The hardware side publishes the mount TF, from the same files **[Proposed]**	Car: `car.launch.py` from `config/car/*.yaml`. Simulator: `polimi_sim` from the same files via `config_dir`. One file, two readers. (R7)
DEC-10	`odom`→`base_link` has one owner at a time **[Proposed]**	The odometry publishes it until the EKF lands (OI-08); then the EKF does, and `publish_tf` goes off on car and simulator in the same change. Two publishers make the tree jump. (R8)
DEC-11	Everything runs in Docker, code bind-mounted **[Decided]**	Same images on Mac, PC and car; the host needs only Docker and git.
DEC-12	In a fork, upstream's files take only upstreamable fixes; our code lives in `polimi_*` packages, in our repo or in a fork's own folder **[Decided]**	Upstream fixes can be pulled and offered back, one commit each; only fixes we cannot make from outside touch upstream's files. `polimi_sim` lives in the simulator fork's `polimi_sim/` folder, never mixed with upstream's files (RASD NFR-MNT-001).

ID	Decision	Rationale / alternatives
DEC-13	JetPack 7.2.1 on the Orin Nano **[Decided]**	Jetson Linux 39.2.1, Ubuntu 24.04 like the `ros:jazzy` containers, CUDA 13: the GPU is usable from the container (RASD C-024), for example by a learned line detector (OI-06). JetPack 6 is Ubuntu 22.04, where a 24.04 container runs on the CPU only. (8 Nov 2026.)

3 Data

The measured car lives in `ros2_ws/config/`, written once and read by both the car and the simulator. The simulator fork's defaults only fill gaps and never hold our measurements; it logs where each value came from.

File	Readers	Holds
`config/car/laser_mount.yaml`	car bringup, simulator	`base_link`→`laser`: today f1tenth's defaults 0.27, 0, 0.11 m, [MEASURE] on the car
`config/car/camera_mount.yaml` **[TBD]**	car bringup, simulator	`base_link`→`camera_link`
`config/lidar/sl450.yaml`	simulator	SL450 model: 40 Hz, 270°, 0.2°, ranges 0.05–25 m, noise, dropout
`config/maps/*.yaml`	simulator	track maps for the simulator
`maps/`	SLAM, AMCL	saved room maps
`polimi_bringup/config/mux.yaml` **[Proposed]**	mux	priorities, timeouts (0.2 s), locks
`polimi_bringup/config/` slam, amcl, ekf **[Proposed]**	logic	node parameters
`vesc.yaml` (`f1tenth_stack`)	VESC driver	gains, servo offset, wheelbase
`joy_teleop.yaml` (`f1tenth_stack`)	joystick teleop	button map: LB enables `/teleop`; RB's `autonomous_control` gates nothing today (OI-09)

4 Interfaces

4.1 The car interface

This table is the contract the stack is built against; it becomes the README of `polimi_bringup` **[Proposed]**. All message types are standard ROS ones, so no interface package is needed. Exact camera and IMU topic names follow `orbbec_camera`'s defaults, so the car needs no remaps **[TBD: T-02]**.

Only `/ackermann_cmd` goes from the stack to the hardware; every other row comes from the hardware.

Topic / frame	Type	Car	Simulator
`/ackermann_cmd`	`AckermannDriveStamped`	`ackermann_to_vesc`	actuation model
`/teleop`	`AckermannDriveStamped`	`joy_teleop`	keyboard bridge
`/joy`	`Joy`	`joy_node`	stand-in **[TBD: T-01]**

Topic / frame	Type	Car	Simulator
`/scan`, frame `laser`	`LaserScan`	`orbbec_camera` LiDAR	LiDAR model
`/odom`	`Odometry`	`vesc_to_odom`	odometry model
TF `odom`→`base_link`	TF	`vesc_to_odom`	odometry model
camera image, info	`Image`, `CameraInfo`	`orbbec_camera`	camera model (`TDD_Gym`)
IMU	`Imu`	`orbbec_camera`	IMU model (`TDD_Gym`, deferred until OI-08)
TF `base_link`→`laser`, `camera_link`	static TF	`car.launch.py`	`polimi_sim`

`/drive` (`AckermannDriveStamped`) is inside the stack, not part of the car interface: any controller writes it, and it enters the mux at priority 10.

4.2 Path following

- **Input path:** `nav_msgs/Path` on one topic, any frame; offline paths latched, online paths recomputed per camera frame (FR-PTH-002, 005).
- **Input pose:** TF lookup from the path's frame to `base_link` (FR-PTH-003).
- **Path stamp:** an online path carries the `header.stamp` of the image it came from, so the pose lookup and the simulator's line-detection score (`TDD_Gym`) use the image time, not the processing time (FR-LIN-002).
- **Output:** `/drive`. A stale online path stops the car (FR-PTH-007).
- **Controllers:** the PID and Pure Pursuit, interchangeable (FR-PTH-004). Paths are closed loops only (FR-PTH-005).
- **Tuning:** each controller publishes its lateral error, heading error and look-ahead point as topics for plotting; every gain is a live parameter (FR-PTH-009).

5 Security & Privacy

- **Physical safety comes first:** the mux, its locks and AEB (FR-SAF) are the safety design; they run on the car and in the simulator unchanged (DEC-03).
- **Network:** on Mac and PC the container's ports bind to 127.0.0.1 only. The car's container uses host networking; ROS traffic on the car's network is not authenticated **[TBD]**.
- **Secrets and personal data:** none in the stack. Camera images may show people in the room; nothing is recorded by default.
- **Repository access:** protected branches and required reviews, as in `CONTRIBUTING.md`.

6 Requirements Traceability

Requirement	Component(s)	Test(s)
FR-DRV-001	`vesc_driver`, `vesc_ackermann`	bench: `ros2 topic echo /odom` while driving by joystick
FR-DRV-002	`joy`, `joy_teleop`	bench: `/teleop` only while the enable button is held
FR-DRV-003	`orbbec_camera`	bench: rates of `/scan`, image, IMU

Requirement	Component(s)	Test(s)
FR-DRV-004	`car.launch.py` (DEC-09)	`tf2_echo base_link laser` matches `laser_mount.yaml`
FR-DRV-005	calibration runbook in `docs/software/`	the procedure run on the car; its result in `camera_mount.yaml`
FR-MAP-001, 002	`slam_toolbox`	AC-V1-01
FR-LOC-001, 002	AMCL (default, OI-07)	AC-V1-01
FR-LOC-003	EKF (DEC-10)	**[TBD: OI-08]**
FR-LIN-001, 002	`polimi_lines`	recorded camera data; AC-V1-02
FR-PTH-001 to 009	`polimi_control`	unit tests on the error and window search; error topics while driving; AC-V1-02, 03
FR-SAF-001 to 005	`ackermann_mux`, `mux.yaml`, `polimi_safety`	AC-V1-04, in the simulator first
FR-SIM-001 to 003	car interface, `hw:=sim`	`ros2 topic info` on both sides; AC-V1-02
FR-DOC-001	runbooks, RASD, TDD	review by someone outside the team
FR-ADV-001 to 003	**[V2]** **[TBD: OI-01]**	AC-V2-A
FR-ADV-011 to 013	**[V2]** **[TBD: OI-01]**	AC-V2-B
NFR-POR-001	images, DEC-01, DEC-11	the same launch on `arm`, `x86`, `jet`
NFR-VAR-001, 002	logic	laps on more than one layout and room
NFR-VAR-003	`polimi_lines`	**[TBD: OI-06]**, once floor and tape are known
NFR-PER-001	logic, controller	laps at 3 m/s (AC-V1-02)
NFR-PER-002	`orbbec_camera` LiDAR	bench: `ros2 topic hz /scan`
NFR-PER-003	`orbbec_camera` camera	bench: `ros2 topic hz` on the image, $\geq$ 30 fps; 60 fps once 1280×720 at 60 in YUYV is verified
NFR-PER-004	`orbbec_camera`, `vesc_driver`	**[TBD]**: method defined with the car's measurement session
NFR-PER-005	`mux.yaml`, VESC	stop sending `/drive`: the mux drops it after 0.2 s and the VESC timeout stops the motor (0.5 s in the simulator, the VESC app value on the car) [VERIFY]
NFR-SAF-001	mux, locks, AEB	AC-V1-04, in the simulator and on the car
NFR-SAF-002	test plan	AC-V1-04 passed on the car before autonomous laps
NFR-MNT-001, 002	development standards (DEC-12)	PR review
FR-OFF-001	**[TBD: OI-11]**	optional

7 Implementation

7.1 Technology Stack

Layer	Choice	Why
Compute	Jetson Orin Nano Super, 500 GB NVMe	BOM; fits the F1TENTH deck
OS, container	Ubuntu 24.04 images, Docker; JetPack 7.2.1 on the car, GPU through the NVIDIA Container Toolkit **[TBD: T-10]**	DEC-01, DEC-11, DEC-13
Middleware	ROS 2 Jazzy, `rmw_zenoh_cpp`	DEC-01
Drivers	`f1tenth_system` (`vesc`, `ackermann_mux`, `teleop_tools`), `orbbec_camera`	C-022
Mapping, localization	`slam_toolbox`, `nav2_amcl`, `robot_localization`	scope; DEC-02
Visualization	Foxglove bridge; RViz through noVNC	runs without a display on the host
Python	system Python of the image; `uv` on the host	house rule

7.2 Plan

1. **Jazzy migration** (section 7.3): Jazzy images, the JAX gym, the simulator fork as a submodule.
2. **`polimi_bringup`:** the car interface README, `stack.launch.py`, `sim.launch.py`, `common.launch.py`, our `mux.yaml`; `simcar` calls `hw:=sim`; with DEC-08 the mux leaves the simulator fork.
3. **Safety:** OI-09 locks and AEB, with the `/joy` stand-in (T-01), tested in the simulator.
4. **Simulator camera** (`TDD_Gym`): the room world with a taped track, the drawn camera and `track_monitor`, for AC-V1-02 in the simulator. Bag replay is built earlier, on today's images.
5. **Camera on the bench**, before the car is built: the 335L with `orbbec_camera`, its topic names (T-02) and profiles (`TDD_Gym` S-06), the tape and floor chosen (RASD A-002, A-003; `TDD_Gym` S-04), the first bags of the real tape.
6. **Logic:** mapping and localization, line detection (on replayed camera bags of the real tape and on the drawn camera, `TDD_Gym`), path following (PID and Pure Pursuit, error topics), as the base platform needs.
7. **Car bringup**, when the car is built: `car.launch.py`, udev rules, measured configs, the camera calibration procedure, safety tests on the car.
8. **Advanced topic** [V2], once OI-01 is decided.

7.3 Jazzy migration

Moves every image to ROS 2 Jazzy, the simulator to the JAX gym, and `polimi_sim` into the simulator fork. Four steps, each checked before the next; the distro, the gym and the frames change in separate commits, so a failure points to one change.

1. **Image:** every image on `ros:jazzy`, `f1tenth_system` on `jazzy-polimi`, the DEC-02 packages, and the same gym and bridge as today, so a breakage is about ROS only. Checks: the images build for any host UID (Ubuntu 24.04 images ship a user with UID 1000); `cv2` and NumPy load from the gym's pip copies (DEC-02); the packages and drivers build; today's simulator runs.

2. **Simulator in the fork:** `polimi_sim` moves into `f1tenth_gym_polimi` with its history and is ported to the JAX gym and the `dev-jazzy` bridge, with labelled default configs and a config overlay (`TDD_Gym`). The mux stays in the fork until DEC-08.
3. **Submodule:** the fork becomes `ros2_ws/src/f1tenth_gym_ros`, `polimi_sim` leaves this repo, and the measured car configs stay in `ros2_ws/config/`. Steps 2 and 3 merge together. Checks: the workspace layout; on the new gym, the scan matches the gym's beam by beam and physics keeps $\geq$ 95 % of 100 Hz on the Mac; the config overlay logs each file's source.
4. **GitHub:** default branches and rules move to the Jazzy branches; the Humble images and volumes are removed after acceptance.

Rollback: the Humble images, volumes and branches stay until step 4, so going back is a revert and the old image tags.

7.4 Development Standards

- **Branches and reviews:** every change through a PR. Protected branches (`main`; the forks' `humble-*`, `jazzy-*`, `dev-humble*`, `dev-jazzy*`) block force pushes and deletion and need one approval.
- **Submodules:** changing a fork takes a commit there, then a pointer commit here; `push.recurseSubmodules=check` blocks a pointer to an unpushed commit. `scripts/setup_git.sh` sets this up and adds the `upstream` remotes.
- **Upstream sync:** merge upstream into our branch, never rewrite it; preview with `git log ours..upstream`.
- **Pins:** gym and bridge refs are bumped on purpose, never by following a branch (NFR-MNT-002).
- **Rebuilds:** code and configs need only `cb`; a `package.xml`, the Dockerfile or a pin needs an image rebuild.
- **Python:** `uv` on the host, never `pip install` directly.

8 Testing

8.1 Strategy

Level	Scope	Tooling / gate
Unit	Controller math, line fitting, AEB time to collision	`colcon test`, pytest
Contract	The car interface on both sides	`ros2 topic info` and `tf2_echo`: same names, types, frames on `hw:=car` and `hw:=sim`
System, simulator	Full flows: map, localize, laps, safety	AC-V1-01 to 04 in the simulator first
System, car	The same flows on the car	AC-V1-01 to 04 on the car; safety before autonomous laps (NFR-SAF-002)

8.2 Acceptance Criteria

As in `RASD_F1tenth`: AC-V1-01 to 04 for the base platform, AC-V2-A or AC-V2-B for the advanced topic. Leaving the lines is judged by the people watching on the car, and measured in the simulator.

9 Open Items

ID	Topic	Notes
T-01	Deadman and e-stop in the simulator	Docker Desktop on macOS has no USB passthrough, so there is no joystick in the container. The locks need a stand-in for `/joy`: a keyboard or Foxglove key that holds the deadman and toggles the emergency stop, built with the OI-09 locks (Plan, step 3).
T-02	Camera and IMU topic names [VERIFY]	From `orbbec_camera`'s Gemini launch, once the camera is on the bench; the simulator's camera model uses the same names.
T-03	LiDAR frame name [VERIFY]	From `orbbec_camera`'s `lidar.launch.py`; set it to `laser` in `car.launch.py` if it differs.
T-04	Confirm DEC-03 to DEC-10	DEC-08 is decided to follow the Jazzy migration (8 Nov 2026). The others are still Proposed; with DEC-06, `RASD_F1tenth` C-022 is reworded.
T-05	Wheelbase	`vesc.yaml` has 0.25 m, the simulated odometry the gym's 0.3302 m. Measure the car and align both.
T-06	Mount measurements [MEASURE]	`laser_mount.yaml` holds f1tenth's defaults; the camera mount comes from the calibration procedure (FR-DRV-005).
T-07	VESC device	udev rule on the Jetson for `/dev/sensors/vesc`, expected by `vesc.yaml`.
T-08	`orbbec_camera` version	2.9.3 is in the Jazzy apt repo; check it with the SL450's firmware at car bringup.
T-09	Unmatched remap	Today's bringup remaps `ackermann_cmd_out`→`ackermann_drive`, which matches nothing: the mux publishes `ackermann_cmd`. Gone with DEC-06.
T-10	CUDA in the container [VERIFY]	With JetPack 7.2.1 (DEC-13), check at bringup how CUDA reaches the `ros:jazzy` container through the NVIDIA Container Toolkit: mounted from the host, or installed in the `jet` image.
T-11	Battery connector [BOM GAP]	The VESC has XT90; the BOM has no TRX (Traxxas High-Current) to XT90 adapter. Check the Zeee 3S packs' connector (section A.3).
T-12	Steering limits [MEASURE]	`vesc.yaml`'s servo clamp gives +0.31 rad left and −0.26 rad right (minimum turning radius about 1.0 / 1.2 m); the simulator clamps at ±0.4189 rad (`TDD_Gym`). Measure on the car; the simulator then takes the car's limits.

A Hardware Selection

The trade studies behind the picks in `RASD_F1tenth` C-010, C-012 and C-013. Prices are the ones seen on the date of each study; the master BOM holds the current vendors and prices.

A.1 LiDAR (study of 23 Sep 2026)

> **Pick: Orbbec Pulsar SL450** (C-012). For F1TENTH the scan rate matters far more than range: at 8 m/s a 10 Hz LiDAR lets the car move 80 cm between scans, which makes scan matching and localization much harder. The SL450 is the only candidate that matches the Hokuyo UST-10LX, the standard F1TENTH LiDAR (40 Hz, 0.25°, 270°, about EUR 1,200).

The study was written for a track bounded by walls. The track is now tape on the floor (RASD C-001 to C-005): the LiDAR cannot see it and maps the room, so "track" about range or SLAM reads "room". The pick is unchanged.

Model	Price	Rate	Res.	Range 70–90% / 10%	FOV	Weight	Interface, principle
Orbbec Pulsar SL450	EUR 531	**40 Hz**	0.2°	45 / 15 m	270°	320 g	Ethernet; dToF, 72 kHz, IP65
Richbeam Lakibeam 1L	EUR 549	30 Hz	0.25°	40 / 20 m	270°	160 g	Ethernet, USB-C; dToF, 45 kHz, 9–36 V
Richbeam Lakibeam 1	EUR 439	30 Hz	0.25°	25 / 15 m	270°	160 g	Ethernet, USB-C; dToF, 43.2 kHz, 9–36 V
Richbeam Lakibeam 1S	EUR 340	20 Hz	0.5°	15 / 10 m	270°	160 g	Ethernet, USB-C; dToF, 6–36 V
Slamtec RPLIDAR S3	EUR 549	20 Hz	0.225°	40 / 15 m	360°	115 g	UART 1 Mbps; dToF, 5 V
LSLIDAR N301	EUR 1,100	20 Hz	0.36°	30 m / –	360°	406 g	Ethernet; ToF, IP67
Slamtec RPLIDAR A2M12	EUR 227	15 Hz	∼0.225°	12 m / –	360°	190 g	UART 256 kbps; triangulation

All seven from RobotShop EU, with a ROS 2 driver; accuracy ±2–3 cm for the ToF units, 1–2.5% of range for the A2M12.

Scan rate	At 5 m/s	At 8 m/s
10 Hz	50 cm	80 cm
20 Hz	25 cm	40 cm
30 Hz	17 cm	27 cm
40 Hz	12.5 cm	20 cm

- **SL450:** the only one at 40 Hz (rates 15, 20, 25, 30, 40 Hz), and 0.2° even at 40 Hz thanks to its 72 kHz sample rate; multi-echo noise filtering. Against: 320 g, twice the Lakibeams, and new (March 2025), so less community troubleshooting than Slamtec.
- **Lakibeam 1, the fallback:** 30 Hz at 0.25° for EUR 439, 160 g, runs straight from a 3S LiPo; Richbeam's ROS 2 driver publishes `/scan`. The 1L only adds range we do not need.
- **Rejected for speed:** Lakibeam 1S (0.5° at 20 Hz), RPLIDAR S3 (capped at 20 Hz), LSLIDAR N301 (twice the price, the heaviest, 20 Hz), RPLIDAR A2M12 (15 Hz, 12 m, slow UART).
- **Field of view:** 360° gives nothing on an F1TENTH car: the chassis and electronics block the rear, so 270° is the norm.

A.2 Camera (study of 7 Oct 2026)

> **Pick: Orbbec Gemini 335L** (C-013), EUR 420 from OpenElab, DigiKey Italy at EUR 408 as a second source. The Intel RealSense D435i stays in the BOM as an unselected alternative.

The LiDAR cannot see the tape, so a camera finds the lines (RASD FR-LIN). What that asks of the camera, most important first:

Need	Why
Global-shutter colour	At 5–8 m/s a rolling shutter skews and smears the lines, worst in corners. The tape colour is not decided, so colour matters.
Wide field of view, $\geq$ 60 fps	Both lines stay in view near the car and through corners. At 8 m/s, 30 fps is 27 cm per frame.
Accurate pixel-to-floor projection	Good factory intrinsics, and the camera's pitch and roll to the floor at every frame: braking and accelerating pitch the car, so a fixed mount calibration is not enough at speed.
Depth aligned to colour, and an IMU	To measure that pitch and roll each frame: fit the floor plane in the depth image, or use the IMU.
Hardware timestamps	10 ms of timestamp error at 8 m/s is 8 cm (NFR-PER-004).
USB 3, light	The Orin Nano's only Ethernet port goes to the LiDAR; a 1/10 car.

#	Camera	Price	Notes
1	Orbbec Gemini 335L / 336L	EUR 420 (335L)	Global-shutter colour 1280×800 up to 60 fps, 94°×68°; depth aligned to colour in the camera; IMU up to 1 kHz; depth from 0.17 m, 95 mm baseline; 133 g, IP65. The 336L adds an IR-pass filter on the depth cameras.
2	RealSense D436	EUR 576	The same capabilities (1280×800 at 60, 90°×65°, aligned depth, IMU) in the D435i's body, same driver and topics.
3	Arducam B0385 (OV9782) plus an IMU board	EUR 69 + ∼EUR 25	Global-shutter colour only: no depth, no factory calibration, no hardware IMU sync. USB 2, so 100 fps only as MJPEG.
4	Luxonis OAK-D W	≈EUR 522	Global-shutter colour, 127°, can run a segmentation network on the camera. Strong fisheye distortion, weaker depth, rare in F1TENTH.
5	RealSense D435i	EUR 405	Rolling-shutter colour, 1080p at 30 fps, 69°×42°. Usable only if the tape shows on its global-shutter IR cameras.

- **Skipped:** Gemini 335/336 (short baseline), OAK-D Lite, OAK-D S2 (rolling-shutter colour); ZED Mini and ZED 2i (rolling shutter, depth on the Orin's GPU; the global-shutter ZED X needs a GMSL2 card); RealSense D455 (bigger, from EUR 536, minimum depth 0.4–0.5 m misses the floor in front of the car).
- **Colour-camera offset:** the colour camera sits beside the left IR camera, the depth origin: 23.75 mm along x on the 335L; the IMU is at (7.866, 1.068, −14.248) mm. Both are factory

transforms and the floor calibration absorbs them. Mount the colour lens on the car's centreline.

- **Projection:** fit a floor plane to the depth image rather than read depth at each tape pixel; stereo depth on a floor seen at a shallow angle is noisy.
- **335L over 336L:** the 336L's IR-pass filter darkens the IR images under LED room light with the projector off, so the 335L is safer if those images are ever used alone.

A.3 Chassis (study of 24 Sep 2026)

> **Pick: Traxxas Ford Fiesta ST Rally VXL, 74276-4** (C-010), bought complete (Plan A). It looks end of life [VERIFY]: Traxxas shows no price and UK retailers list it as discontinued. Plan B, if no complete car can be found, is a rolling chassis from Traxxas spare parts.

The F1TENTH (RoboRacer) build guide is written for the Traxxas Slash 4x4 Premium (brushed Titan motor, XL-5 ESC). Applied to the Rally VXL's parts list (REV. 260209):

Guide step (Slash 4x4)	On the Rally VXL
Remove the body clips and body	Clipless body (7420, 7414): body and trim discarded.
Remove the XL-5 ESC	Remove the VXL-3s ESC (3355R); the VESC 6 MkVI replaces it.
Remove the receiver box, receiver, antenna tube	7424, TQi receiver 6533, 1726 and 1926 discarded: the joystick and the Jetson replace the radio.
Replace the brushed motor	Not needed: the Velineon 3500 brushless (3351R) is kept; the BOM's 4 mm-to-3.5 mm bullet adapters connect it to the VESC.
Keep the servo	The 2075 servo is kept; the VESC drives it through the PPM cable.
Remove the battery hold-down	7426 discarded; the battery sits under the platform deck.
Remove the nerf bars, reuse their M3 screws	No nerf bars on the Rally [VERIFY]; take M3 screws from the BOM's kit.
Three 45 mm M3 standoffs into the chassis	Chassis 7422 ("Slash LCG and Rally"): check the hole pattern on the real part [VERIFY].
Battery to VESC through TRX to XT90	Not in the BOM **[TBD: T-11]**.

Kept (the rolling chassis, motor and servo): chassis tub, bulkheads, tie bars, skidplate, gear cover; the 4WD driveline (sealed front and rear diffs, centre slipper clutch, centre driveshaft, half-shafts, spur and pinion, motor mount and plate); the suspension (arms, pins, caster and steering blocks, stub-axle carriers, shock towers, camber and toe links, Ultra Shocks); the steering (bellcranks, servo saver, drag link, toe links, 2075 servo); wheels, tyres, hubs, bearings; the Velineon 3500.

Discarded: body and accessories (7427-x, 7420, 7414, 7410, 7413X, 7419); receiver box, seals and plug (7424, 7425, 3698); battery hold-down (7426); antenna tube (1726, 1926); VXL-3s ESC, BEC and fan (3355R, 2260, 3340); TQi radio and telemetry (6509R, 6528, 6533); the box's batteries and tools.

A.3.1 Plan B parts

Universal Traxxas part numbers, sold in EU shops under the same number (prefix TRX or TXX). [VERIFY] marks a variant or colour not confirmed as stock; check them, and every quantity,

against the exploded views of the 74276-4.

Qty	Part	Description	Notes
Chassis and structure			
1	7422	Chassis, Slash LCG / Rally	platform-deck standoffs go here [VERIFY]
1	7430X	Bulkhead, front	
1	7429	Bulkhead, rear	
1	6823	Bulkhead tie bars, front and rear, aluminium	colour [VERIFY]
1	7435	Front skidplate, angled	
1	7477	Gear cover with wire retainer	guards the spur gear against cables
1	7434	Foam body bumper, low profile	recommended
Driveline			
1	6788	Front differential, complete (13/37, rally)	
1	6789	Rear differential, complete (13/37, rally)	
1	6878	Slipper clutch, complete	a centre slipper, not a centre diff
1	6893R	Slipper input shaft, bearing adapter, pin	may come with 6878 [VERIFY]
1	6888	Centre rear drive hub	
1	6888X	Front drive hub, hardened steel	
1	7455	Centre driveshaft, plastic	[VERIFY]; 6855 aluminium is the upgrade
2	7450	Front half-shaft, Extreme HD, 6 mm axle	sold as left or right
2	7451	Rear half-shaft, Extreme HD, 6 mm axle	sold as left or right
1	6842	Spur gear, 0.8 metric pitch	stock tooth count **[TBD]** (50, 52 or 54T)
1	–	Pinion, 32-pitch steel, 3 mm bore	stock tooth count **[TBD]**; the smaller stock pinion suits indoor speeds
1	7460A	Motor mount, 3x6 flat-head screw, nylock nut	
1	7490	Motor plate	colour [VERIFY]
Suspension			
2	6731	Suspension arms, front/rear L&R (2 per pack)	4 arms in total
1	9042	Suspension pin set, Extreme HD	9042X is the hardened upgrade
1	9032	Caster blocks, Extreme HD	colour [VERIFY]
1	9037	Steering blocks, Extreme HD, L&R	colour [VERIFY]
1	9050	Rear stub-axle carriers, Extreme HD	colour [VERIFY]
1	9038	Front shock tower, Extreme HD	
1	9039	Rear shock tower, Extreme HD	
1	7432	Camber links, 49 mm (72 mm c-c)	front/rear [VERIFY]
1	7431	Camber links, 49 mm (63 mm c-c)	front/rear [VERIFY]
1	7433	Toe links, 47 mm (77 mm c-c), rear	[VERIFY]
1	3760A	Ultra Shocks, long, with springs (front pair)	[VERIFY]; rebuild kit 2362
1	3762A	Ultra Shocks, XX-long, with springs (rear pair)	[VERIFY]
1	5529	Shim set, hollow balls	
1	2742	Rod ends, long, hollow balls	
Steering			

Qty	Part	Description	Notes
1	2075	Digital high-torque waterproof servo	kept by the build
1	6845X	Bellcranks, servo saver, spring, retainer, horn	
1	7438	Drag link	6845A aluminium is the upgrade
1	7439	Front toe links, composite, hollow balls	
1	2545	Bellcrank bushings, 5x8x2.5 mm	
Wheels and hubs			
2	7473	Tyres and wheels, glued, gravel (2 per pack)	gravel grips poorly indoors; road tyres **[TBD]**
1	9069	Wheel hubs, 12 mm hex, Extreme HD	
1	2754	Stub-axle pins	
1	3647	Wheel nuts, 4 mm flanged nylock	[VERIFY]; 1747 is the alternative
1	5854	Hub retainer, 17 mm, M4	used on this model [VERIFY]
Motor			
1	3351R	Velineon 3500 brushless, 3.5 mm bullets	kept; the 3350R combo is easier to find, then drop its ESC

- **Bearings** (blue rubber-sealed, 2 per pack), quantities **[TBD]** from the exploded views: 5116 (5x11x4 mm), 5117 (6x12x4), 5118 (8x16x5), 5119 (10x15x4), 5120 (12x18x4), 1985 PTFE washers (5x8x0.5).
- **Screws and nuts** (hex drive, packs of 6), quantities **[TBD]**: button-head 3x6 to 3x40 (2575–2580, 2582, 2583, 2591, 2592) and 4x10/12/14 (3936–3938); cap-head 2.5x6, 2.5x8, 3x8, 3x15 (3215, 3965, 1552, 2586); flat-head 3x6, 3x8, 3x15 (3932, 3931, 3646); countersunk 3x6, 3x12, 4x12, 4x15 (2534, 2552, 2542, 2546); nylock nuts 3 and 4 mm (2745, 1747); grub screws (2743); washers 3x6 (2746); shock shoulder screws (3642X). One missing screw stops the build: this is Plan B's biggest risk.
- **Tools:** spare parts come without the box's hex keys; add a hex driver set (Traxxas 2748R or 8712, or 1.5/2.0/2.5 mm hex and 4/7 mm nut drivers).
- **Cost:** Plan B heads toward or above the price of the complete car before the motor, servo, diffs, shocks, bearings and screws are priced, and it adds a full assembly. Plan A stays the choice.